Three Unifications: A Falsifiable Map of the Quantum-Gravity Frontier

Complete paper (English edition)

MuningAn · PlanetarySystem · 2026-08

Three Unifications: A Falsifiable Map of the Quantum-Gravity Frontier

MuningAn[1] · **PlanetarySystem**[1]

[1] PlanetarySystem. Correspondence: MuningAn. Repository: github.com/everest-an/Antigravity

Manuscript type: Article (conceptual, theory). **Draft:** 1.0, 2026-08.

Abstract

"Unified field theory" conflates three logically distinct claims, and the conflation has produced a century of confusion in both folk physics and popular science. Here we separate the claims and assign each its own evidential regime. Transformational unification, the claim that one field physically converts into another, is quantitatively falsified for laboratory electromagnetic sources: Earth-scale gravity would require field strengths roughly eight times the Schwinger limit, at which the vacuum becomes unstable. Geometrical unification, the claim that known forces are projections of a higher-dimensional geometry, remains open but confined: its gravitational Aharonov-Bohm signatures are excluded at the electronvolt scale by orbital fifth-force bounds and testable only in the nanoelectronvolt window now accessible to the Th-229 nuclear clock. Algebraic unification, the claim that gauge theory and gravity share a deeper kinematic algebra, is an established mathematical fact that is silent on ontology. Building on this classification, we prove a control-channel uniqueness theorem: within general relativity plus the Standard Model, any engineering operation preserving all sources, the metric structure, and the interaction channels leaves all geodesic motion unchanged, so that inertia engineering has no degrees of freedom independent of stress-energy engineering. We organize the nineteen live propositions of the field into a decision matrix in which every row carries a discriminating experiment, a target theory it would exclude, and a timescale; sixteen rows are anchored by reproducible computations, and nine representative computations are summarized. The result is a descriptive map rather than a new theory: it explains where a century of folk proposals failed, and it identifies the three live candidates for the source variable of spacetime, each with a scheduled experiment.

Keywords: unified field theory; double copy; quantum gravity; gravitationally induced entanglement; equivalence principle; falsifiability; nuclear clock

1. Introduction

The phrase "unified field theory" has been used for at least three different intellectual projects, and their conflation is the root of a century of confusion. The first project seeks a physical conversion: a process that turns one field into another, as in the folk claim that changing electromagnetic fields generate gravity. The second seeks a geometrical embedding: a higher-dimensional geometry whose projections include the known forces, the route opened by Kaluza and Klein. The third seeks an algebraic identity: a kinematic structure from which the amplitudes of two different theories both descend, the route now realized by the double copy. These three projects have different evidence bases, fail in different ways, and require different kinds of experiments to adjudicate. A blanket dismissal of "unified field theory" is therefore as uninformative as a blanket endorsement.

This paper makes three contributions. First, we formalize the three-way classification and show that the modern frontier, from the double copy program to tabletop tests of gravitationally induced entanglement, is organized naturally by it. Second, we state and prove a control-channel uniqueness theorem (Section 6) that identifies stress-energy as the only engineering entry point within general relativity plus the Standard Model, with a corollary that quantifies the efficacy of any proposed mass lever. Third, we compile the nineteen live propositions of the field into a decision matrix (Section 7) in which every row carries a discriminating experiment, a target it would exclude, and a timescale, and we anchor sixteen of the rows with reproducible computations summarized in Section 8.

The relation of this work to the existing literature is stated explicitly in Section 9. In brief: the classification has no direct precedent in the preprint literature; the control-channel theorem generalizes the warp-drive no-go results of Bobrick and Martire, of Le, and of Barzegar, Buchert, and Vigneron, which are special cases of its assumptions; and the term "metric engineering" is inherited from Puthoff, whose exploratory register we replace with a specific verdict.

2. The three unifications

We define the three claims as follows.

Definition 1 (transformational unification). Transformational unification asserts a physical map from a field A to a field B such that engineering A produces B as a matter of dynamics, without introducing further structure.

Definition 2 (geometrical unification). Geometrical unification asserts an embedding of A and B as components of a single geometric object, such that both are recovered by projection, with no claim that A produces B.

Definition 3 (algebraic unification). Algebraic unification asserts a shared algebraic or kinematic structure from which the amplitudes of two theories are obtained by a fixed rule, again with no conversion involved.

The three claims fail in different modes. Transformational unification fails quantitatively: it is a dynamical claim, and its numbers can be computed. Geometrical unification fails, when it fails, by confinement: its distinctive signatures are driven into an ever-narrower window by successive experiments. Algebraic unification does not fail at all as mathematics; what it lacks is an ontological verdict, and no finite computation can supply one. These three failure modes, not the three claims themselves, are the organizing principle of the paper.

3. Transformational unification: quantitative falsification

The cleanest test case of transformational unification is the claim that a changing electromagnetic field generates gravity. The relevant physics is not in dispute: the electromagnetic stress-energy tensor enters Einstein's equations with its full content, energy density and pressure. The only question is magnitude.

A parallel-plate capacitor with a field of 10 MV m^{-1}, of volume one cubic meter, carries an energy density of order 440 J m^{-3} and sources a gravitational acceleration of order 10^{-25} m s^{-2}, twenty-five orders of magnitude below Earth gravity. Reversing the question is more informative. The field strength required to produce a target acceleration g over a characteristic scale L follows from inverting the Poisson equation with the electromagnetic energy density as source:

$$E_{needed} = (2 \rho_{needed} / \varepsilon_0)^{(1/2)}, \quad \rho_{needed} = g c^2 / (8 \pi G L), \quad (1)$$

where the factor of two records the pressure content of the electromagnetic stress-energy tensor. For g = 9.8 m s^{-2} and L = 1 m, the required field is 1.1×10^{19} V m^{-1}, roughly eight times the Schwinger limit $E_S = m_e^2 c^3/(e\hbar) \approx 1.3 \times 10^{18}$ V m^{-1}, the scale at which the quantum vacuum becomes unstable against spontaneous pair production. At that point the program has changed meaning: one is manufacturing matter, not engineering gravity. Equation (1) closes the transformational program for laboratory electromagnetic sources without reference to any speculative physics.

The same arithmetic bounds the adjacent fifth-force window. A Yukawa modification of Newtonian gravity,

$$a_Y(r) = \alpha (G M / r^2) (1 + r/\lambda) e^{(-r/\lambda)}, \quad (2)$$

with coupling α relative to G and range λ, is excluded at order unity across all probed scales. The strongest bounds vary with range: of order 3×10^{-3} at λ = 10 μm from Eöt-Wash torsion balances, 10^{-6} from millimeters to a hundred kilometers, and 10^{-9} to 10^{-15} at orbital and lunar scales. The only remaining opening lies below ten micrometers, where the Casimir force dominates and dedicated platforms are required. A representative landscape appears in Figure 2.

The relation to the warp-drive no-go literature should be stated precisely. Bobrick and Martire showed that any warp drive is a shell of matter moving inertially and therefore requires propulsion. Le showed that steering a positive-energy drive requires radiation and mass loss, $-\dot{m} \geq 3m|\alpha|$. Barzegar, Buchert, and Vigneron classified warp-drive spacetimes and proved new no-go theorems. Each of these results is a special case of the classification we give in Section 6: each shows that a specific metric-engineering proposal cannot bypass the source channel. Our theorem generalizes the logic to arbitrary engineering operations.

4. Geometrical unification: the known window

Geometrical unification is the paradigm case of a claim that is neither established nor excluded, but whose testable content has become precise. An extra compact dimension contributes a tower of graviton modes whose lightest state produces a Yukawa-type correction to the gravitational potential,

$$V(r) = -\left(G m_1 m_2 / r \right) \left(1 + \alpha e^{(-r/\lambda)} \right), \quad (3)$$

where α encodes the coupling of the lightest extra mode relative to standard gravity and λ its Compton range. The gravitational Aharonov-Bohm proposals of the past two years convert this correction into a concrete spectroscopic prediction: a quantum system in free fall around a gravitating body acquires an energy-level splitting

$$\Delta E = m_{sys}\, \alpha\, \Delta\Phi(r, \lambda), \quad (4)$$

where $\Delta\Phi$ is the peak-to-peak variation of the Yukawa-corrected potential over the orbit. Inverting Eq. (4) shows that electronvolt-scale splittings claimed for nuclear systems require α of order 0.4, and millielectronvolt splittings for atomic systems require α of order 10^{-3}. Both values are excluded by four to eight orders of magnitude by the orbital bounds of Section 3. The allowed window therefore sits at the nanoelectronvolt scale.

The instrument for that window is now essentially complete. The Th-229 nuclear isomer transition has been directly laser-excited at 148.18 ± 0.42 nm with a 447 ± 25 s half-life in a VUV-transparent crystal; solid-state hosts with lifetimes exceeding six hundred seconds exist; and a spinless host crystal, $Th(SO_4)_2$, removes the dominant magnetic-dipole broadening channel, with a projected instability of $4.6 \times 10^{-23}/\sqrt{\tau}$. Geometrical unification is no longer a vague hope. It is a specific measurement program with a known instrument and a known energy window.

5. Algebraic unification: mathematics without ontology

The double copy states that gravitational scattering amplitudes can be obtained from the square of gauge-theory amplitudes under color-kinematics duality. Its status has progressed from a computational technique to an algebraic fact through three results. Off-shell N=8 supergravity in four dimensions is realized as the double copy of N=4 super Yang-Mills at cubic order in the fields. Amplitudes on coherent-state gauge backgrounds double-copy to amplitudes on curved spacetimes whose metric is built from the gauge background; in the self-dual sector the map relates exact vacuum solutions. And the Ehlers transformation of general relativity is the double copy of electromagnetic duality, a fact of direct historical resonance with the transformational ambitions of the Tesla era.

What these theorems do not establish is a unified ontology. The residual gauge algebra of the double-copied Schwarzschild solution collapses from an infinite-dimensional structure to the finite-dimensional isometry group, demonstrating that shared algebraic structure does not imply shared physical content. The correct statement is that algebraic unification is real and that its ontological import is undecidable by computation alone. This boundary is a feature of the program's honest assessment, not a defect of the mathematics.

6. The control-channel uniqueness theorem

Theorem 1 (control-channel uniqueness). Let (M, g) be a general-relativistic spacetime with g determined by given sources and boundary conditions. Let E be an engineering operation satisfying:

(A1) **Source invariance:** E does not change any field source in (M, g), including $T_{\mu\nu}$, electromagnetic currents, and all Standard Model source terms;

(A2) **Metric non-intervention:** E does not replace g by a solution g′ not determined by the sources of (A1) and the same boundary conditions;

(A3) **Channel closure:** E introduces no interaction channel beyond the Standard Model.

Then E leaves the geodesic motion of every test particle in (M, g) unchanged.

Proof. The geodesic equation contains the Christoffel symbols, which are determined by g. The metric g is determined by the field equations from the sources of (A1) and the boundary conditions. Assumptions (A2) and (A3) preserve the equations and their solution. Hence the Christoffel symbols are unchanged, and the equations of motion, and therefore their solutions, are unchanged. ■

The content of the theorem is the completeness of (A1)–(A3): they exhaust the control channels available inside general relativity plus the Standard Model. Violating (A1) is stress-energy engineering, which is energy engineering by another name. Violating (A2) is metric engineering, which requires exotic matter or modified gravity. Violating (A3) is new physics, bounded by the fifth-force and Lorentz-violation searches of Sections 3 and 7.

Corollary 1 (inertia engineering). Inertial mass equals gravitational mass (the weak equivalence principle), and general relativity couples to matter only through $T_{\mu\nu}$. Hence inertia engineering has no degrees of freedom independent of stress-energy engineering.

Corollary 2 (mass-lever efficacy). For any decomposition of mass, $m = m_{Higgs} + m_{QCD} + m_{vac} + \ldots$, the gravitational efficacy of each component as an engineering lever equals $\partial T_{\mu\nu}/\partial$(that component). The Higgs enters roughly nine percent of the proton mass term; QCD carries roughly ninety-one percent; and the vacuum contribution is not well-defined, since it depends on the regularization scheme (Section 8).

Corollary 3 (effective mass). Negative effective mass in metamaterials is a property of dispersion relations and does not change $T_{\mu\nu}$; it therefore has no direct gravitational consequence.

Boundary statement. The theorem holds for classical general relativity with classical sources. Quantum superposed sources, emergent-geometry hypotheses, and hybrid classical-quantum models lie outside it. Those exclusions are precisely the open rows of the decision matrix, and we do not claim the theorem applies to them.

7. The decision matrix

We compile the nineteen live propositions of the field into a decision matrix. Every row carries an evidence grade, a discriminating channel, a target theory it would exclude, and a timescale. The full matrix is given in Figure 1; Table 1 gives a representative subset.

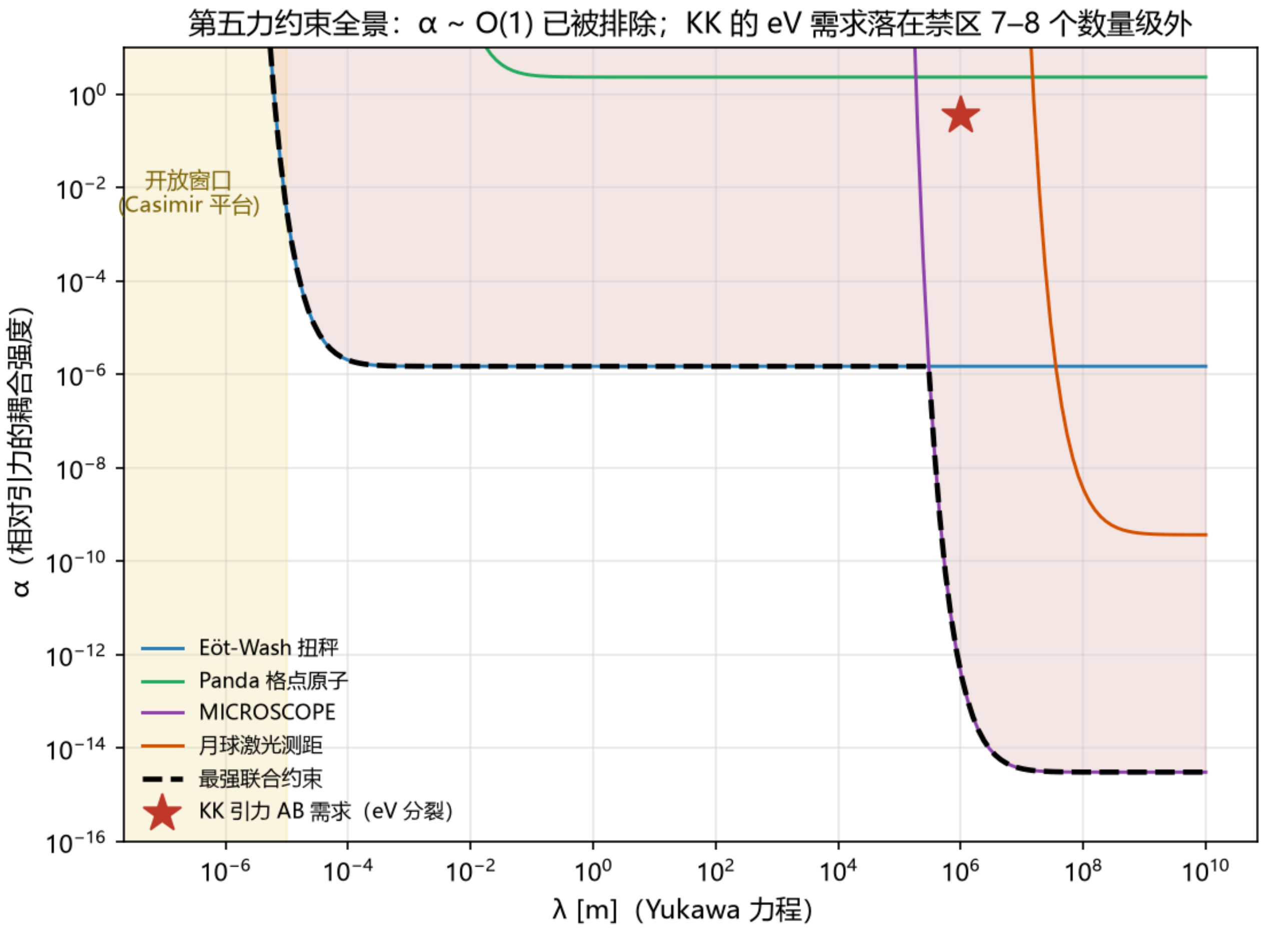
Table 1. Representative subset of the decision matrix.

#	Arrow	Status	Discriminating channel
3	EM fields convert into gravity	falsified	Schwinger-limit computation, Eq. (1)
5	$\langle \hat{T}_{\mu\nu} \rangle$ as the quantum source of geometry	falsified	MEP (Fedida-Kent) + post-Newtonian consistency
9	ICO implies quantum spacetime	falsified	QC-QC theorem (Salzger-Vilasini)
14	Vacuum energy as an engineering resource	falsified	regularization-scheme dependence
1–2	Electromagnetic and electroweak unification	established (experiment)	—
17	Inertia engineering = $T_{\mu\nu}$ engineering	classification statement (Thm. 1)	—
4	Double copy ontology	grade B, undecidable by computation	—

#	Arrow	Status	Discriminating channel
6–7	Superposed sources; GIE witnesses nonclassical gravity	grade B	QGEM; LOCC exclusion
11	Quantum weak equivalence principle	grade B	in-orbit WEP; operator formalism
19	Planck-scale Lorentz violation / GUP	grade B, no significant evidence	GRB/AGN, LLR, optomechanics
8	Post-Newtonian GIE (frame dragging)	grade C, ~7 orders below readout	quantum clock interferometry
12	KK extra dimensions (AB splitting)	grade C; eV closed, neV open	orbital bounds + Th-229 clock
15	Geontropic spacetime fluctuations	grade C; falsifiable at strong scale only	GQuEST
18	Entanglement → geometry (real universe)	grade C	holographic/analog platforms

The methodological content of the matrix is its discipline. A row is admitted only if it states what observation would exclude which theory, and on what timescale. This is the same discipline the 2025 debate over gravitationally induced entanglement taught: an observation discriminates between theories only if no competing theory reproduces it. The debate, in which a classical-plus-matter construction was claimed to reproduce the effect and then shown to carry the entanglement in the matter sector, is discussed in Section 9.


判决矩阵：19 个「统一/反重力」箭头的当前状态（2026-08）
3 EM 场 → 引力（转换式统一）
Schwinger 极限核算
5 $\hat{T}_{\mu\nu}$ 作为量子几何源
MEP + PN 一致性
9 ICO → 量子时空
QC-QC 定理
13 屏蔽第五力
边界已量化
14 真空能可随意弯曲时空
正则化方案依赖
1 电与磁统一
Maxwell / 相对论
2 电弱统一
BEH 机制
17 惯性工程 = Tμν 工程
控制通道唯一性定理
4 double copy 本体论
不可直接判决
6 叠加质量 → 叠加几何
QGEM / 相位读出
7 GIE 见证非经典引力
LOCC 排除
11 量子 WEP
系统误差主导
19 LIV / GUP（Planck 对称破缺）
GRB/LLR/核钟
8 后牛顿 GIE（frame dragging）
缺口 ~7 个数量级
10 因果结构量子化
10 年+
12 KK 额外维度
neV 窗口
15 geontropic 时空涨落
仅强档可证伪
16 引力子量子噪声
bolometric 10 年+
18 纠缠 → 几何（真实宇宙）
无近期通道
不成立
(已排除)
成立 / 已确立
(A 级)
B 级
(数学对应/原理清晰)
C 级
(候选/远期)
每行携带：判决通道 + 判决对象 + 时间尺度（详见 实验路线图.md）
不成立（已判）
成立 / 已确立
B 级
C 级



8. Methods: reproducible anchors

All quantitative claims of this paper are anchored in a companion repository containing nine order-of-magnitude calculations, seven simulations, and five figure-generation scripts, all executable with a single runner. Representative anchors follow.

Entanglement phase. The entanglement phase of the standard quantum-gravity-entanglement benchmark, with two masses m separated by d, each in a spatial superposition of splitting Δx evolved for time t, is

$$\varphi = (G m^2 t / \hbar) (1/(d+\Delta x) + 1/(d-\Delta x) - 2/d), \quad (5)$$

which evaluates to 0.217 rad for the published benchmark parameters. Realistic chip parameters give 10^{-6}–10^{-5} rad, converting a two-hundred-measurement certification into a 10^{9}-to-10^{21}-measurement campaign. A simulation of the protocol in a four-dimensional Hilbert space shows negativity $N \approx 0.078$ and CHSH violation 0.024 for the quantum model, and identically zero for the mean-field and LOCC models, implementing the exclusion logic of row 6–7. A continuous-variable verification in a truncated Fock space reproduces the qubit result with deviation proportional to $e^{-2\alpha^2}$, where α is the branch displacement, confirming the fidelity of the abstraction.

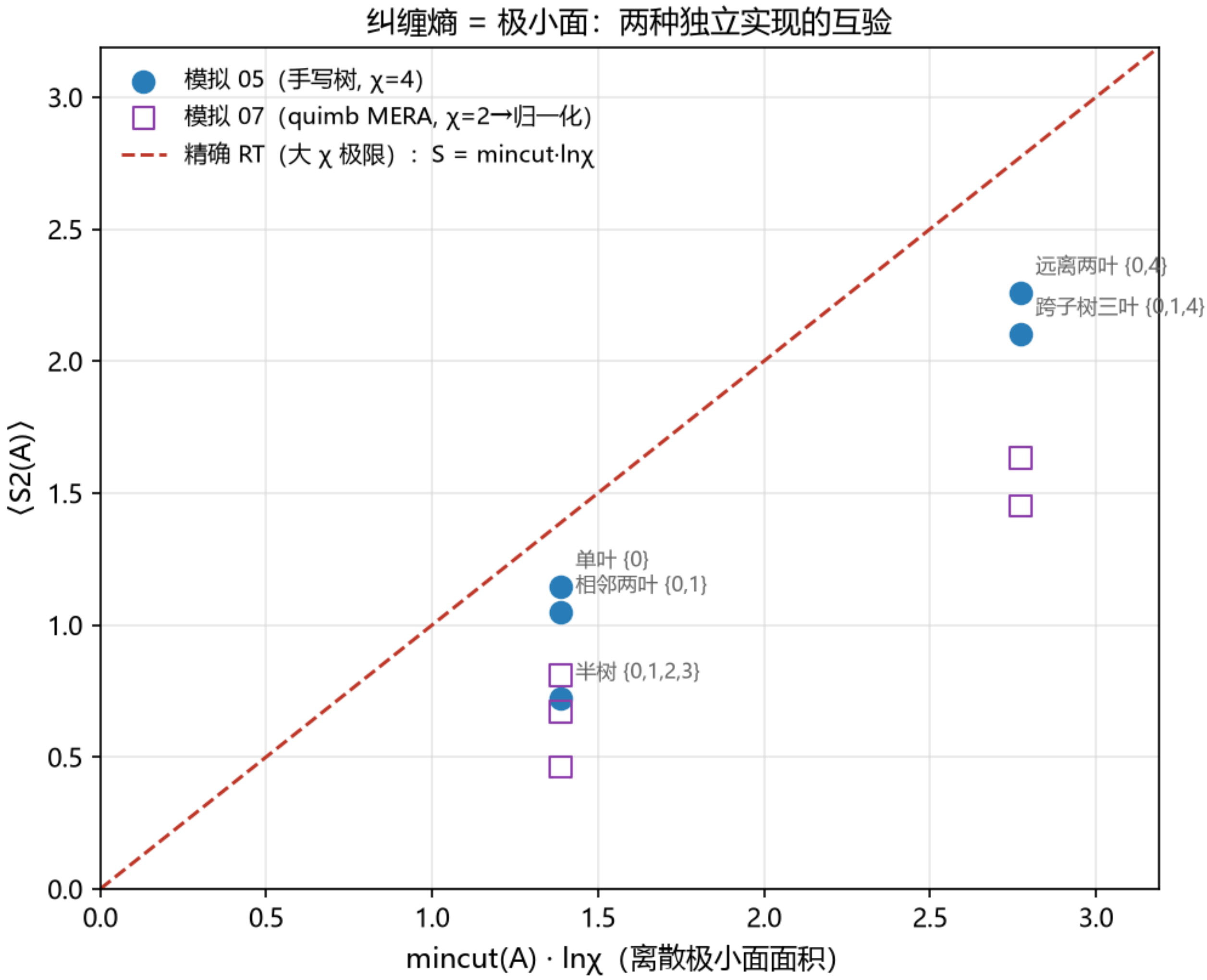
Spacetime fluctuations. For a tabletop Michelson interferometer of arm length L = 5 m with a one-gram mirror at 10 Hz, the free-mass standard quantum limit is 5.2×10^{-18} m Hz^(−1/2). The geontropic benchmark $h = \sqrt{(l_p/L)}$ gives a displacement of 9.0×10^{-18} m, requiring an integration time of about 3 s at the SQL and 0.03 s with photon-counting readout; the weaker benchmark $h = l_p/L$ requires 10^{33} s and is unfalsifiable in practice.

Frame dragging. The Lense-Thirring precession field of a rotating source with angular momentum J is $\Omega\_LT = (G/c^2r^3)[3\hat{r}(\hat{r}\cdot\hat{J}) - \hat{J}]$. Laboratory rotors ($J \sim 10^2$ kg m² s⁻¹) produce surface precessions of order 10^{-23} rad s^{-1}, ten orders below Earth's; the Gravity Probe B measurement of 37 mas yr^{-1} is reproduced to order of magnitude from this formula.

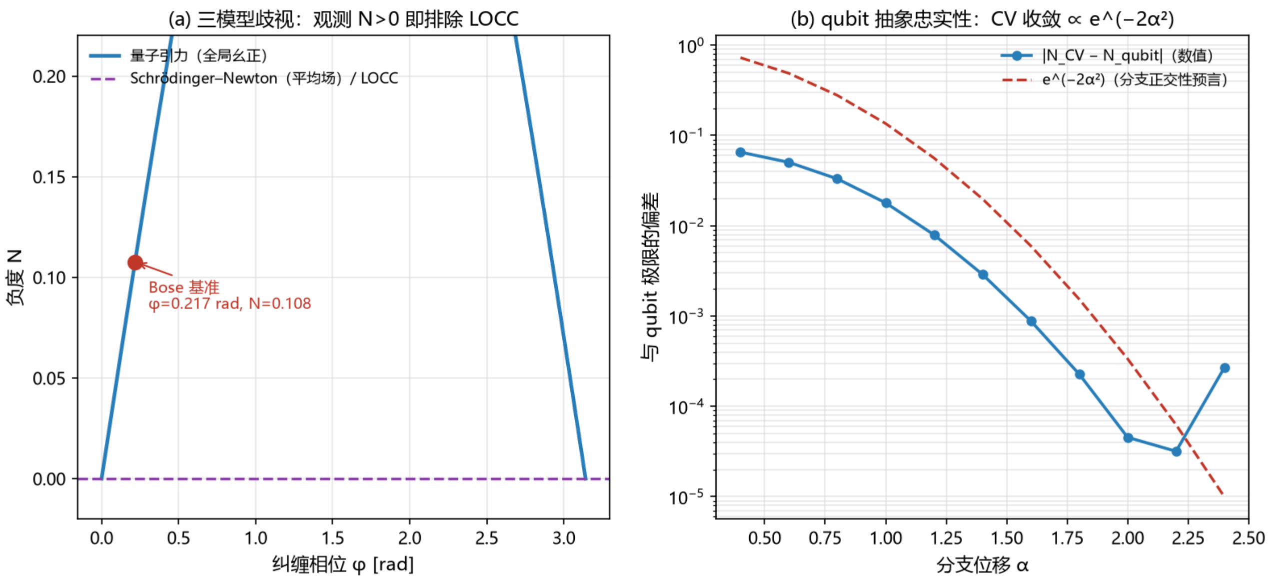
Entanglement and geometry. An isometric tree tensor network of bond dimension χ reproduces $S_2(A) \approx \text{mincut}(A)\cdot\ln\chi$ for boundary intervals, including the disconnected minimal surface for distant intervals; two independent implementations agree.

Vacuum regularization. The vacuum energy density of a free scalar field of mass m is $+\Lambda^4/(16\pi^2)$ under a momentum cutoff Λ, zero under zeta-function regularization in the massless limit, and $m^4/(64\pi^2)[\ln(m^2/\mu^2) + c]$ under dimensional regularization, changing sign with the scale μ. Only differences of vacuum energy, the Casimir type, are scheme-independent.

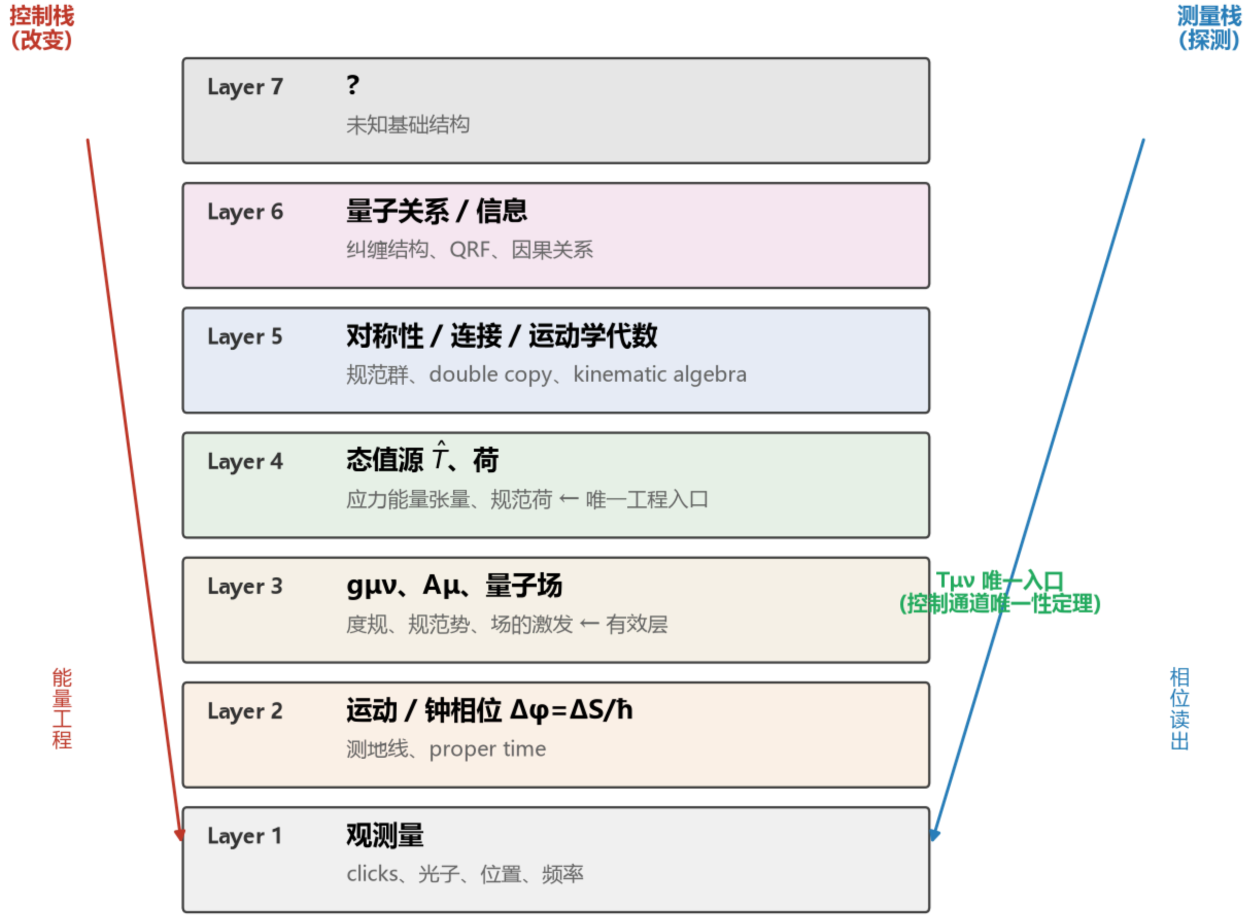
Code availability. All scripts are available at github.com/everest-an/Antigravity.



纠缠熵 = 极小面：两种独立实现的互验
模拟 05（手写树, χ=4）
模拟 07（quimb MERA, χ=2→归一化）
精确 RT（大 χ 极限）：S = mincut·lnχ
远离两叶 {0,4}
跨子树三叶 {0,1,4}
单叶 {0}
相邻两叶 {0,1}
半树 {0,1,2,3}
⟨S2(A)⟩
mincut(A) · lnχ（离散极小面面积）
0.0
0.5
1.0
1.5
2.0
2.5
3.0



(a) 三模型歧视：观测 N>0 即排除 LOCC
量子引力（全局幺正）
Schrödinger–Newton（平均场）/ LOCC
Bose 基准
φ=0.217 rad, N=0.108
负度 N
纠缠相位 φ [rad]
(b) qubit 抽象忠实性：CV 收敛 ∝ e^(−2α²)
|N_CV − N_qubit|（数值）
e^(−2α²)（分支正交性预言）
与 qubit 极限的偏差
分支位移 α



控制栈：人类在'测量栈'领先、在'控制栈'几乎空白——这正是'Gravity is the wrong engineering variable'的含义

9. Discussion

Relation to the GIE debate. The 2025 exchange over gravitationally induced entanglement is the methodological centerpiece of the field's current state. The original argument, that entanglement generated solely through gravity implies a nonclassical mediator, was challenged by a construction in which classical gravity coupled to quantum field theory matter reproduces the effect, and the rebuttal located the entangling channel in the matter sector: the offending diagram vanishes when the two interferometers use distinct, non-interconverting matter species. The lesson we draw is not that the original argument fails, but that its premises must be stated: GIE is a witness against local classical channels, not against every classical model, since Newtonian gravity with classical time evolution generates the same entanglement. Our matrix therefore assigns to row 6–7 the precise target it excludes.

The status of negative energy. Row 14 rests on the regularization dependence demonstrated in Section 8, but the adjacent question of quantum inequalities deserves a calibrated statement. The inequalities restrict negative energy for inertial observers, and their experimental status is contested: a meta-analysis of squeezed-light data finds violations of a proposed inequality.

Macroscopic utilization of negative energy remains excluded regardless; the contestation concerns the status of the bounds, not the availability of the resource.

Limitations. The classification is descriptive, not predictive: it predicts no new phenomenon. The control-channel theorem is a classification statement, and its assumptions are its content; we do not claim it excludes quantum superposed sources, emergent geometry, or hybrid models, which are the open rows of the matrix. The three-way classification is offered as a physics-organizing device, not as a contribution to the philosophy of reduction.

10. Conclusion

A century of unification talk has conflated three claims. Separating them yields a cleaner map than the field has previously had. Transformational unification is closed for laboratory sources by two lines of arithmetic. Geometrical unification is confined to a nanoelectronvolt window with a completed instrument. Algebraic unification is established mathematics with an undecidable ontology. The control-channel theorem then identifies $T_{\mu\nu}$ as the only engineering entry point inside general relativity plus the Standard Model, and the decision matrix schedules the nineteen live propositions against concrete experiments.

The ambition behind the folklore survives this analysis in exactly one form. The question is no longer whether one field can be converted into another. It is what variable, if any, controls spacetime at its source, and the three live candidates, the quantum state, the kinematic algebra, and the relational structure, each have a scheduled experiment. If a technology of spacetime ever exists, it will operate on those candidates, not on a force. The map is the deliverable, and the map is falsifiable, which is all that a scientific claim is allowed to be.

References

1. Tesla, N. Prepared statements and collected writings (primary sources).
2. Zhang, X. Unified Field Theory, v7.2 (folk corpus, primary source).
3. Bose, S. et al. Spin entanglement witness for quantum gravity. *Phys. Rev. Lett.* **119**, 240401 (2017).
4. Marchese, M. M., Plávala, M., Kleinmann, M. & Nimmrichter, S. Newton's laws of motion generating gravity-mediated entanglement. *Phys. Rev. A* **111**, 042202 (2025).
5. Fedida, S. & Kent, A. Mixture equivalence principles and post-quantum theories of gravity. *Phys. Rev. D* **111**, 126016 (2025).
6. Salzger, M. & Vilasini, V. Higher-order quantum processes respecting closed labs in a spacetime. arXiv:2605.08351 (2026).
7. Vermeulen, S. M. et al. Photon counting interferometry to detect geontropic spacetime fluctuations with GQuEST. *Phys. Rev. X* **15**, 011034 (2025).

8. Derevianko, A., Elwell, R. & Hudson, E. R. Colloquium: Nuclear clocks. arXiv:2606.11048 (2026).
9. Hiraki, T. et al. Controlling ^{229}Th isomeric state population in a VUV transparent crystal. *Nat. Commun.* **15**, 5536 (2024).
10. Morgan, H. W. T. et al. A spinless crystal for a high-performance solid-state ^{229}Th nuclear clock. arXiv:2503.11374 (2025).
11. Bonezzi, R., Casale, G. & Hohm, O. The double copy of maximal supersymmetry in D=4. arXiv:2501.02058 (2025).
12. Ilderton, A. & Lindved, W. Coherent states, background fields, and double copy. arXiv:2505.16852 (2025).
13. Holton, B. Residual symmetries and BRST cohomology of Schwarzschild in the Kerr-Schild double copy. arXiv:2509.24112 (2025).
14. Bobrick, A. & Martire, G. Introducing physical warp drives. *Class. Quantum Grav.* **38**, 105009 (2021).
15. Le, A. T. Steering a warp drive without exotic matter. arXiv:2606.22531 (2026).
16. Barzegar, H., Buchert, T. & Vigneron, Q. General formalism, classification, and demystification of the current warp-drive spacetimes. arXiv:2602.16495 (2026).
17. Puthoff, H. E. Advanced space propulsion based on vacuum (spacetime metric) engineering. *JBIS* **63**, 82 (2010).
18. Maclay, G. J. & Davis, E. W. Testing a quantum inequality with a meta-analysis of data from squeezed light. *Found. Phys.* **49**, 797 (2019).
19. Ford, L. H. & Roman, T. A. Negative energy seen by accelerated observers. *Phys. Rev. D* **87**, 085001 (2013).
20. Wilczek, F. Origins of mass. *Cent. Eur. J. Phys.* **10** (2012).
21. Yang, Y.-B. et al. Proton mass decomposition from the QCD energy momentum tensor. *Phys. Rev. Lett.* **121**, 212001 (2018).
22. Liu, K.-F. Hadrons, superconductor vortices, and cosmological constant. *Phys. Lett. B* (2023).
23. Panda, C. D. et al. Measuring gravity by holding atoms. *Nature* **631**, 515 (2024).
24. Ofengeim, D. D. & Piran, T. The 300 TeV photon from GRB 221009A. *Phys. Rev. D* **112**, 083055 (2025).
25. Du, S.-S. et al. Hierarchical test of Lorentz invariance with GRB spectral-lag measurements. *Astrophys. J.* (2025).
26. Jalalzadeh, S. & Moradpour, H. Finite Hilbert space and maximum mass of Schwarzschild black holes from a GUP. *Phys. Lett. B* (2026).
27. Chiao, R. Y. et al. Gravitational Aharonov-Bohm effect. *Phys. Rev. D* **109**, 064073 (2024).
28. Jusufi, K. et al. Signatures of modified gravity from the gravitational Aharonov-Bohm effect. *J. Cosmol. Astropart. Phys.* (2025).
29. Céleri, L. C., Soares-Pinto, D. O. & Turolla Vanzella, D. A. The apparatus strikes back. arXiv:2607.08819 (2026).

30. Wakakuwa, E. et al. Relativistic gravity-induced entanglement via frame dragging. arXiv:2606.31678 (2026).

Author contributions

MuningAn conceived the classification and the theorem, performed all computations, and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Data availability

All computations, simulations, figures, the full nineteen-row decision matrix, and the complete reference list are available at github.com/everest-an/Antigravity.